

CHEMISTRY

MARCH-2016

General Instructions:

- (1) The question paper has 4 parts. All parts are compulsory.
- (2) Part-A carries 10 marks. Each question carries one marks.
- Part-B carries 10 marks. Each question carries two marks.
- Part-C carries 15 marks. Each question carries three marks.
- Part-D carries 35 marks. Each question carries five marks.
- (3) Write balanced chemical equations and draw neat diagrams wherever necessary.
- (4) Use log tables and simple calculator if necessary. (Use of scientific calculator is not allowed.)

PART-A

I. Answer all the questions. Each question carries one mark. (Answer each question in one word or one sentence): 10×1=10

1. State Raoult's Law for liquid mixtures.

Sol. The vapour pressure of each liquid in the solution is equal to the product of mole fraction of solute and vapour pressure of the pure component.

2. Define molality.

Sol. Molality (m) is defined as the number of moles of solute dissolved per kilogram of the solvent.

3. The rate equation for the reaction $A+B \rightarrow P$ is, $\text{rate} = K[A]^{\frac{1}{2}}[B]^2$. What is the order of reaction?

Sol. $\text{Rate} = [A]^{\frac{1}{2}}[B]^2$

$$\text{Order} = \frac{1}{2} + 2$$

$$\text{Order} = \frac{5}{2} = 2.5$$

4. Write the equation for the anodic reaction during rusting of iron.

Sol. At anode $\rightarrow Fe \rightarrow Fe^{2+} + 2e^{-}$; (oxidation).

5. Name a hormone which maintains the constant level of glucose in the blood.

Sol. Insulin.

6. Give an example for Homogeneous catalysis.

Sol. $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

7. Which gas is liberated when 2-bromopropane is heated with alcoholic potash?

Sol.



8. Mention the main commercial source of helium.

Sol. Natural gas.

9. Give reason: Acetophenone does not react with saturated sodium bisulphite solution.

Sol. Due to steric hindrance.

10. Among carbon and carbon monoxide which one is better reducing agent for Fe_2O_3 above 1000K?

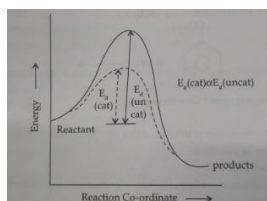
Sol. Above 1000K carbon is better reducing agent for Fe_2O_3 .

PART-B

II. Answer any five of the following. Each question carries two marks. $5 \times 2 = 10$

11. Explain the effect of catalyst on activation energy of a reaction with graph.

Sol.



According to Arrhenius equation, lesser the activation energy (E_a) faster will be rate of reaction. In the presence of catalyst, a reaction proceeds in an alternative pathway with lesser activation energy. Thus, the reaction takes place faster.

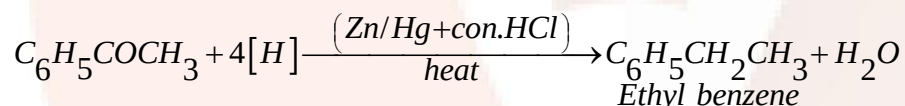
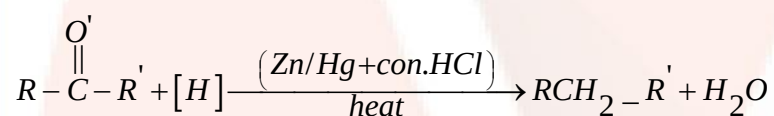
12. What is Schottky defect? What is its effect on the density of a solid?

Sol. Schottky defect: It is basically a vacancy defect in ionic solid. It arises if some of the atoms or ions are missing from their normal lattice sites. In order to maintain electrical neutrality, the number of missing cations and anions are equal.

Schottky defect lowers the density of solid.

13. Calculate the Λ_m^0 for $MgCl_2$. The limiting molar conductivities of Mg^{2+} and Cl^{-1} ions are $106.0 \text{ S cm}^2\text{mol}^{-1}$ and $76.3 \text{ S cm}^2\text{mol}^{-1}$ respectively.

Sol.



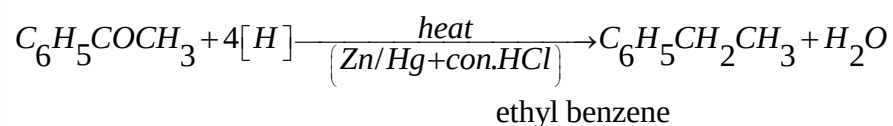
$$\begin{aligned}\Lambda_m^0(MgCl_2) &= \lambda^0_{Mg^{2+}} + 2\lambda^0_{Cl^{-}} \\ &= (106.0 \text{ S cm}^2 + 2 \times 76.3 \text{ S cm}^2)\end{aligned}$$

$$\Lambda_m^0(MgCl_2) = 258.6 \text{ S cm}^2$$

14. Explain Clemmensen reduction with an example.

Sol. Aldehydes and ketones are heated with zinc amalgam and conc. HCl, to form corresponding hydrocarbons. Some fault

Example-



15. What are Analgesics? Give an example for a non-narcotic analgesic.

Sol. Drugs that are employed to relieve pain are called analgesics. Example for non-narcotic analgesic is Aspirin.

16. Give reason for the following:

(i) Actinoids generally show variable oxidation states.

Sol. $5f$ – $6d$ – and $7s$ -subshells of Actinoids have comparable energies. As a result one or more electrons from $5f$ subshell also participates in bonding. Hence actinoids shows variable oxidation state.

(ii) Lanthanoids are less reactive than actinoids.

Sol. Because Lanthanoids have less SRP (standard reduction potential) value than Actinoids, Therefore they are less reactive.

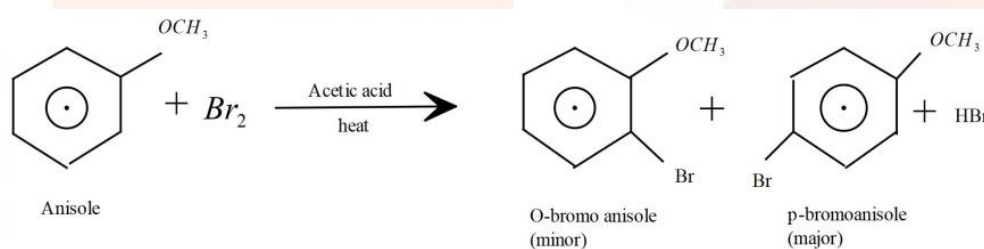
17. What are anionic detergents? Give an example.

Sol. These are defined as the sodium salt of long chain alkyl benzene sulphonic acid.

Example: Sodium Lauryl sulphate ($C_{11}H_{23}CH_2OSO_3Na$).

18. What is the action of bromine in ethanoic acid on anisole? Give equation.

Sol.



PART-C

III. Answer any five of the following. Each question carries 3 marks.

5×3=15

19. Write the equation involved in the manufacture of ammonia by Haber's process and mention optimum conditions for maximum yield of ammonia.

Sol. Chemical equation:
$$N_2(g) + 3H_2(g) \xrightarrow[Al_2O_3]{Fe_2O_3} 2NH_3(g) \quad \Delta H_f = -46.1 KJmol^{-1}$$

Optimum conditions for maximum yield:

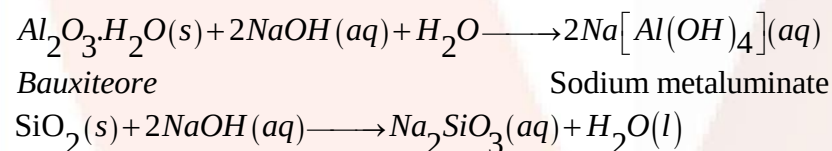
High pressure of 200 atmospheres (Or 200×10^5 Pa)

Optimum temperature of 700K.

Iron oxide is a catalyst mixed with small amount of K_2O or Al_2O_3 .

20. How is pure alumina obtained from bauxite by leaching process?

Sol. Bauxite is the ore of aluminium. It associated with oxides, titanium oxides, and SiO_2 . Powderd ore id digested with 45% aqueous solution caustic soda at 473K-523K and 35-36 bar pressure. Al_2O_3 goes in to solution forming soluble sodium aluminate silica and dissolves as sodium silicate.



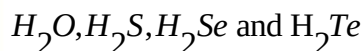
21.(a) How is chlorine prepared in the laboratory using $KMnO_4$?

Sol. In a laboratory chlorine is prepared by the action of conc. HCl on solid $KMnO_4$ Solid.

(b) Give reason: Interhalogen compounds are more reactive than halogens (except fluorine).

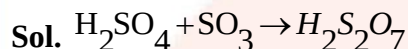
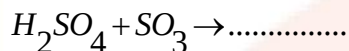
Sol. Typically, interhalogens are more reactive the halogens m because interhalogen bonds are weaker than diatomic halogen bonds except for fluorine. It is due to difference in electronegativity For example, ICl is more reactive than iodine because the electronegativity difference polarize the bond between iodine and chlorine and therefore breaks more easily than iodine bond. Thus, Inter halogen compounds are more reactive than halogens since the bond between two dissimilar halogens atoms is weaker than the bond between two similar halogen atoms.

22.(a) Among the following which one is more acidic? Give reason.



Sol. H_2Te is more acidic, because the size of the central atom increases. The bond (H-Te) dissociation enthalpy in H_2Te is low, therefore the H^+ ion is easily lost.

(b) Complete the following equation:



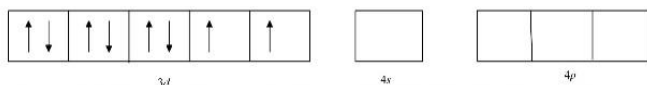
23. Explain the hybridization, geometry and magnetic properties of $[Ni(CN)_4]^{2-}$ ion using valence bond theory. (Z of Ni=28)

Sol.

In the complex $[Ni(CN)_4]^{2-}$, nickel is in +2 Oxidation state and is d^8 ion.

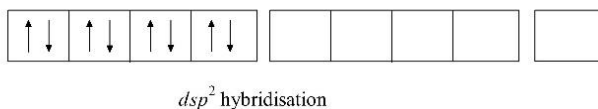
The hybridization scheme is like this,

Ni :



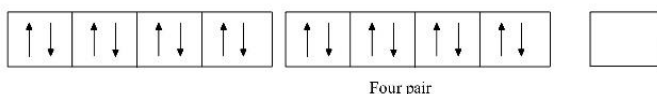
The strong ligand like CN^- pairs up the unpaired electrons in 3d-orbital

Ni :



The empty dsp^2 hybridized orbitals overlap with the filled orbitals, one from each of the CN^- ligand

Ni :



The complex is square planar, and is diamagnetic because all the electrons are paired up in the complex.

24. Explain the preparation of potassium permanganate from MnO_2 .

Sol. Potassium permanganate is prepared by fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 . This results in the formation of the dark green K_2MnO_4 which disproportionates in a neutral or acidic solution to give potassium permanganate.

25. (a) Give reason :

Sc^{3+} ions are colourless whereas V^{3+} ions are coloured.

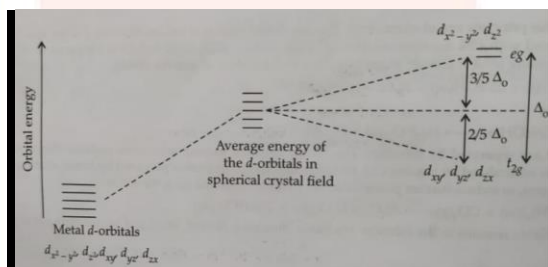
Sol. Sc^{3+} ions are colorless, whereas V^{3+} ions contain partially d-orbitals. Therefore they have d-d transition of electron, which results in imparting color to the ions.

(b) What are interstitial compounds?

Sol. Interstitial compounds are those which are formed when small atoms like H, C and N are trapped inside the crystal lattices of metals.

26. (a) Explain the crystal field splitting of d-orbitals in octahedral complexes.

Sol.



When ligands approach the central metal ion, the electrons in the d-orbitals of the central metal ion are repelled by the lone pairs of the ligands. Because of these interactions, the degeneracy of the d-orbitals of the metal ion is lost and they split into sets of orbitals having different energies. The splitting of d-orbitals of the metal ion is lost and it splits into two sets of orbitals having different energies. The splitting of d-orbitals is shown in the figure.

(b) Write the formula of tetrammine aquachloridocobalt (III) chloride.

Sol. $[Co(NH_3)_4(H_2O)Cl]Cl_2$.

PART-D

IV. Answer any three of the following. Each question carries 5 marks.

27. (a) 200cm^3 of an aqueous solution of a protein contains 1.26 g of protein. The osmotic pressure of such a solution at 300K is found to be 2.57×10^{-3} bar. Calculate the molar mass of protein. ($R = 0.083\text{L bar mol}^{-1}\text{K}^{-1}$)

Sol. The given quantities are.

$$\Pi = 2.57 \times 10^{-3} \text{ bar}$$

$$V = 200\text{cm}^3 = 0.200\text{Litre}$$

$$T = 300\text{K}$$

$$R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$$

$$W_2 = 1.26\text{g}$$

$$M_2 = \frac{W_2 RT}{\Pi V}$$

$$= \frac{1.26\text{g} \times 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1} \times 300\text{K}}{2.57 \times 10^{-3} \text{ bar} \times 0.200\text{Litre}}$$

$$= 61038.91\text{gmol}^{-1}$$

(b) What are non-ideal solutions? Mention the reason for the negative deviation from the Raoul's law

Sol. The solutions which do not obey Raoult's Law over the entire range of concentration and temperature are called as non-ideal solutions.

The reason for the negative deviation from Raoult's law is that the intermolecular attractive forces.

Between A-A & B-B are weaker than those between A-B and leads to decrease in vapour pressure.

28. (a) Resistance of a conductivity cell filled with 0.02M KCl solutions is 520Ω . Calculate the conductivity and molar conductivity of that solution.

[Cell constant of the cell $= 1.29\text{cm}^{-1}$]

Sol. Given,

$$c = 0.02 \text{ mol L}^{-1}$$

$$R = 520 \Omega$$

$$G^* = 1.29 \text{ cm}^{-1}$$

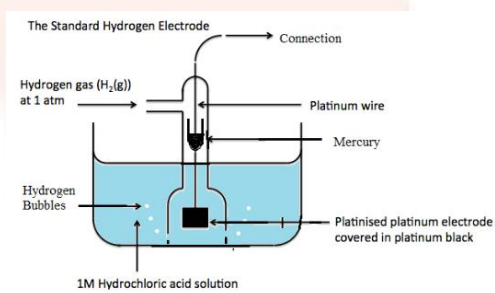
$$\begin{aligned} \text{Conductivity (K)} &= \frac{G^*}{R} = \frac{1.29 \text{ cm}^{-1}}{520} \\ &= 0.248 \times 10^{-2} \text{ S cm}^{-1} \end{aligned}$$

$$\text{Molar conductivity } (\Lambda_m) = K \times 1000 \text{ cm}^3 \text{ L}^{-1} / \text{molarity}$$

$$\begin{aligned} &= \frac{0.248 \times 10^{-2} \text{ S cm}^{-1} \times 1000 \text{ cm}^3 \text{ L}^{-1}}{0.02 \text{ mol L}^{-1}} \\ &= 124 \text{ S cm}^2 \text{ mol}^{-1} \end{aligned}$$

(b) Draw the neat labeled diagram of standard hydrogen electrode.

Sol.



29. (a) Calculate the packing efficiency of simple cubic crystal.

Sol. Edge length or side of a cube = a

Radius of a particle = r

Particles touch each other along the edge.

$$\therefore a = 2r, \text{ Volume of the cell } = a^3 = (2r)^3 = 8r^3$$

Simple cubic unit cell contains only 1 atoms

$$\text{Volume occupied} = \frac{\text{volume of one atom}}{\text{volume of the unit cell}} \times 100\%$$

$$\begin{aligned} &= \frac{\frac{4}{3}\pi r^3}{8r^3} \times 100 \\ &= 52.4\% \end{aligned}$$

(b) An element crystallizes in fcc lattice. If edge length of the unit cell is 408.6 pm and density is 10.5 g cm^{-3} , Calculate the atomic mass of the element. [$N_A = 6.022 \times 10^{23}$]

Sol.

$$\begin{aligned} a &= 408.6 \times 10^{-12} \text{ m} \\ &= 408.6 \times 10^{-10} \text{ cm} \end{aligned}$$

$$d = 10.5 \text{ g cm}^{-3}$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$Z = 4$$

Therefore, the crystal is fcc lattice

$$\begin{aligned} M &= \frac{da^3 N_A}{Z} = \frac{10.5 \text{ g cm}^{-3} \times (408.6 \times 10^{-10} \text{ cm})^3 \times 6.022 \times 10^{23} \text{ mol}^{-1}}{4} \\ &= 107.8 \text{ g mol}^{-1} \end{aligned}$$

* Atomic mass of the element = 107.8u.

30. (a) Give reason for the following:

(i) Brownian movement of colloidal particles.

Sol. The reason for the Brownian movement of colloidal particles is the collisions of the constantly moving dispersion medium particles, colloidal particles move slowly in a zig-zag manner.

(ii) Stability of colloids.

Sol. The reason for the stability of colloids is the presence of equal and the similar charges on the colloidal particles. The repulsive forces between charged particles having same charge prevent them from aggregating when they come closer to one another.

(iii) The movement of colloidal particles towards cathode in an electric field.

Sol. The movement of colloidal particles towards cathode is due to the presence positive charges on the colloidal particles.

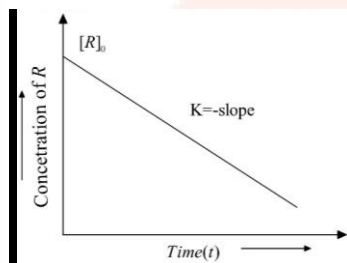
(b) What is peptization? Give an example.

Sol. Peptization is defined as the process of converting a freshly prepared precipitate into colloidal sol by shaking it with dispersion medium in the presence of a small amount of electrolyte.

Example: Freshly precipitate silver iodide is converted into colloidal solution of silver iodide with small amount of silver nitrate or potassium iodide solution.

31. (a) Derive an integrated rate equation for the rate constant of a 1st order reaction.

Sol.



Consider a general first order reaction.



The differential rate equation is

$$\text{Rate} = -\frac{d[R]}{dt} = k[R]$$

Where K, is the rate constant of the first order reaction.

$$\text{On rearranging, } \frac{d[R]}{[R]} = -kdt$$

Integrating on both the sides, we get

$$\ln[R] = -Kt + I \dots \dots \dots (i)$$

Where I is the constant of integration.

When $t=0$, $[R] = [R]_0$ and hence using equation (i)

$$I = \ln[R]_0 \dots \dots \dots (ii)$$

Rearranging this equation.

$$\ln[R]_0 - \ln[R] = Kt$$

$$\Rightarrow K = \frac{1}{t} \ln \frac{[R]_0}{[R]} \dots\dots\dots(iii)$$

$$\Rightarrow K = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \dots\dots\dots(iv)$$

(b) Draw the graph for [R] versus time (t) for a zero order reaction. Give the relationship between the constant and the slope of the curve.

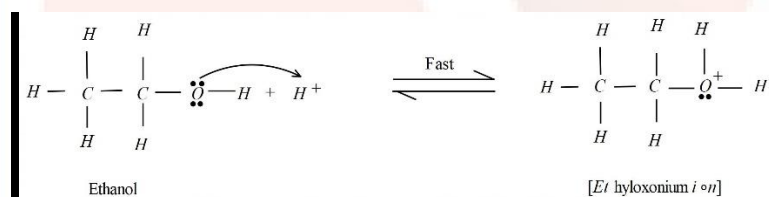
Sol.

FIGURE

V. Answer any four of the following, each question carries five marks. 4×5=20

32. (a) Explain the mechanism of acid catalyzed dehydration of ethanol to ethane.

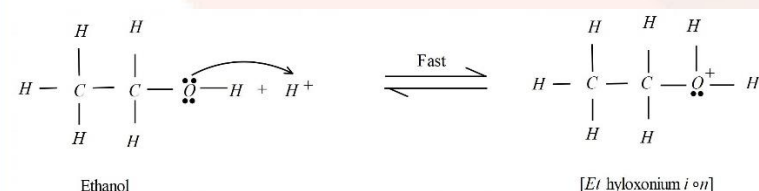
Sol.



The mechanism of acid catalyzed dehydration of ethanol to ethene follows the following steps

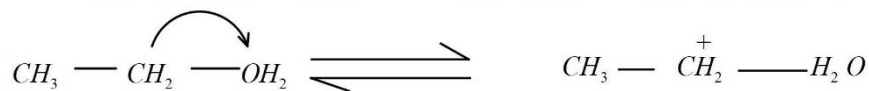
Step-1: Protonation of alcohol.

Concentrated sulphuric acid protonates ethyl alcohol to form ethyloxonium ion (protonated alcohol)



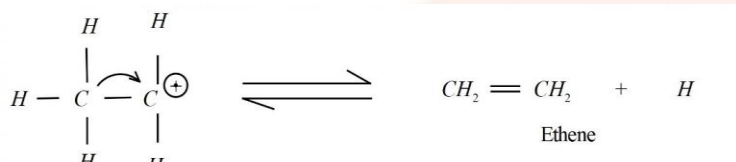
Step-2: Formation of Carbocation

The oxonium ion formed loses stable water molecule to form a carbocation. This is the slowest step of the mechanism hence is the rate determining step.



Step-3: Elimination of proton to form ethane.

The carbocation formed loses a proton to form the final product ethene.



(b) What is the effect of the following groups on the acidity of phenol?

(i) $-\text{CH}_3$ group

Sol.

The presence of $-\text{CH}_3$ group in the phenol decreases the acidity of phenol as it is an electron donating group. Therefore it will intensify the negative charge and thus, release of a proton becomes more difficult.

As a result, the acidity decreases.

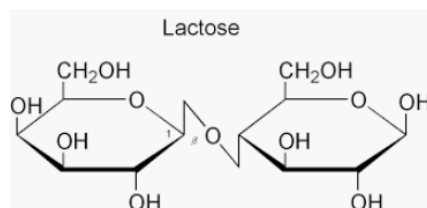
(ii) $-\text{NO}_2$ group

Sol.

Since $-\text{NO}_2$ group is an electron withdrawing group. It disperses, the negative charge of the phenoxide ion and thus stabilizes it. Therefore, NO_2 group will increase the acidic strength of phenols.

33. (a) Write the Haworth's structure for lactose.

Sol.



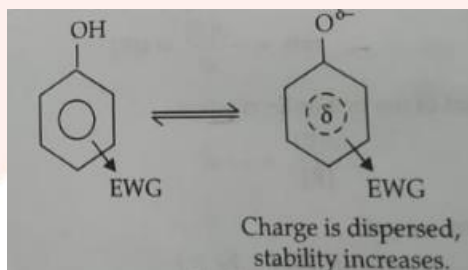
(b) Give an example for the following:

(i) Naturally occurring optically inactive α - amino acid.

Sol. Glycine.

(ii) Nitrogen base only found in R.N.A.

Sol.



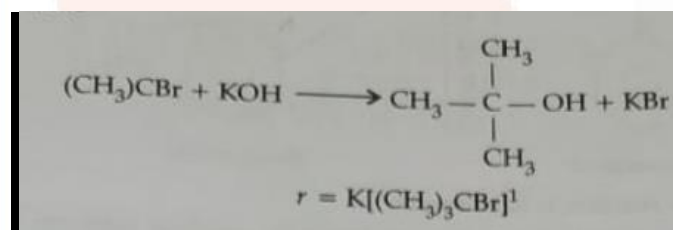
Uracil.

(c) Name the disease caused by the deficiency of vitamin A.

Sol. Night blindness.

34. (a) Explain S_N1 mechanism with an example.

Sol.



S_N1 Reaction

When tertiary butyl bromide is heated with aqueous potassium hydroxide, tertiary butyl alcohol is formed and follows the first order Kinetics. i.e. The rate of reaction depends on the concentration of only one reactant i.e. tertiary butyl bromide.

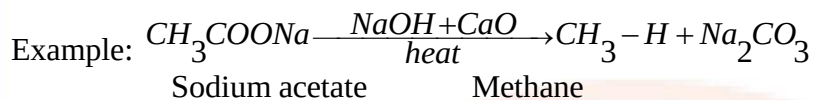
$$r = K[(CH_3)_3CBr]^1$$

Mechanism:

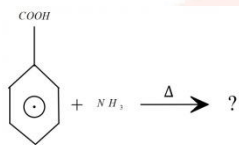
Step-1: Formation of carbocation.

(b) Explain decarboxylation reaction with an example.

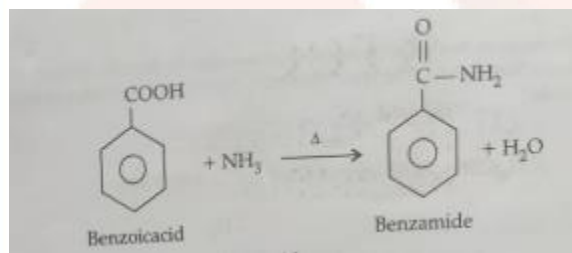
Sol. Carboxylic acid lose carbon dioxide to form hydrocarbons when their sodium salts are heated with soda lime (NaOH and CaO). This reaction is called as decarboxylation.



(c) What is the major product of the following reaction?



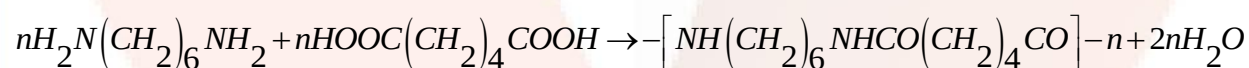
Sol.



The major product of the reaction is benzamide.

36. (a) What are condensation polymers? Give an example.

Sol.



The polymers that are formed by repeated condensation reaction between two different bi-functional or tri-functional monomeric units with the elimination of simple molecules like water, methyl alcohol, ammonia, hydrogen chloride etc are called as condensation polymers.

Example: Nylon-6,6 is formed by the condensation hexamethylene diamine with adipic acid.

(b) Write the following:

(i) IUPAC name for the monomer of natural rubber.

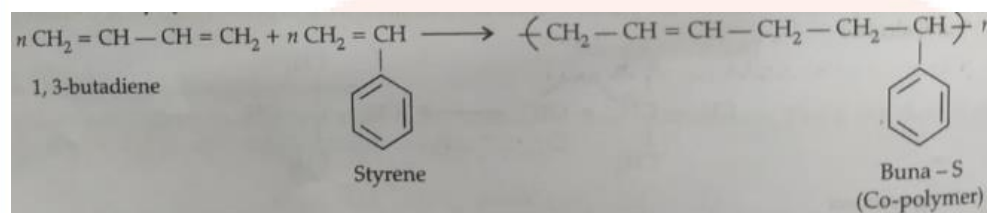
Sol. IUPAC name for the monomer of natural rubber is 2-methyl-1,3-butadiene

(ii) The partial structure of polythene.

Sol. The partial structure of polythene is $\text{-(CH}_2\text{ - CH}_2\text{)-}$

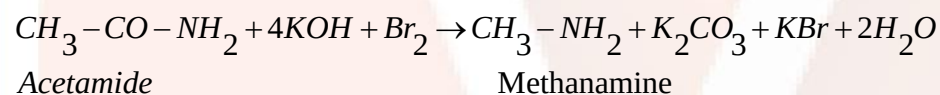
(c) Give an example for a co-polymer.

Sol. The addition polymers that are formed by two different monomeric species are called copolymers.



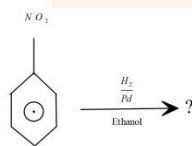
37. (a) How do you prepare methanamine from Hoffmann bromamide degradation reaction?

Sol. When acetamide reacts with bromine and potassium hydroxide at 343K, methanamine is formed

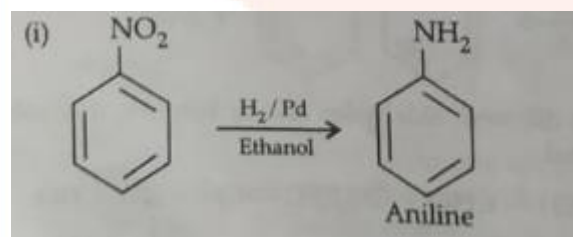


(b) Complete the following reactions:

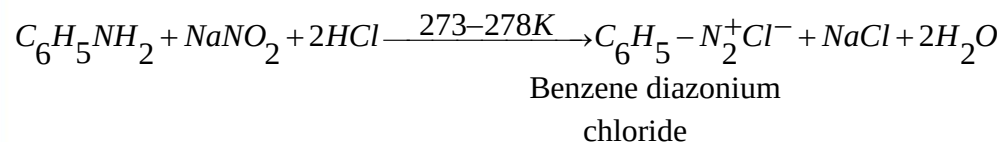
(i)



Sol.



(ii)



(c) Give reason: Ammonia is more basic than aniline.

Sol. Ammonia is more basic than aniline because in aniline the lone pair of electrons on nitrogen are partially withdrawn by benzene ring and it becomes difficult for nitrogen to donate electron pair.